Majorana Modes in the Machine

Block 2: Finite-Size Physics & Edge Modes

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Outline

1. Numerical Method
2. Finite-Size Results
3. Edge Mode Localization
4. Majorana Hybridization Splitting
5. Summary

Step 1: Building the BdG matrix

The real-space BdG Hamiltonian is a $(2L \times 2L)$ block matrix:

$$M = \begin{pmatrix} h & d \\ -d & -h \end{pmatrix}, \quad h_{ij} = -\mu\,\delta_{ij} - t(\delta_{i,j+1} + \delta_{i+1,j}), \quad d_{ij} = \Delta(\delta_{i,j+1} - \delta_{i+1,j})$$

```python
def build_hamiltonian(self):
    h = np.zeros((L, L))
    np.fill_diagonal(h, -mu)            # onsite: chemical potential
    for j in range(L - 1):
        h[j, j+1] = h[j+1, j] = -t    # hopping

    d = np.zeros((L, L))
    for j in range(L - 1):
        d[j, j+1] =  delta              # pairing (antisymmetric)
        d[j+1, j] = -delta

    return np.block([[h, d], [-d, -h]])
```

Step 2: Diagonalization & extracting the spectrum

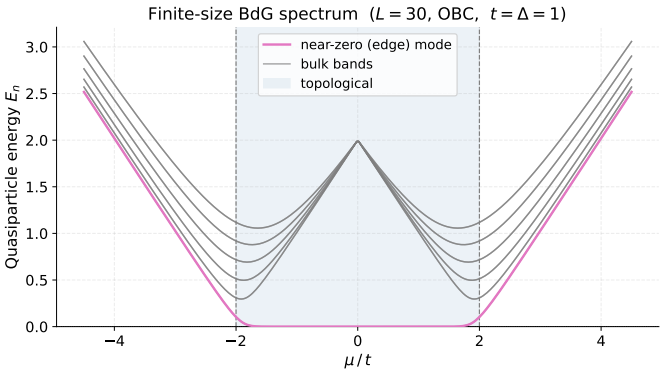
```
def positive_spectrum(self):
    evals = np.linalg.eigvalsh(self.build_hamiltonian())
    return np.sort(np.abs(evals))[:self.L]
```

Why `np.abs` instead of filtering `evals >= 0`?

Particle-hole symmetry guarantees eigenvalues in $\pm E$ pairs. Near-zero Majorana modes can round to tiny *negative* values at double precision — a sign filter silently drops them, causing spurious spikes in the finite-size spectrum at large L. Taking $|E|$ and keeping the L smallest values is exact.

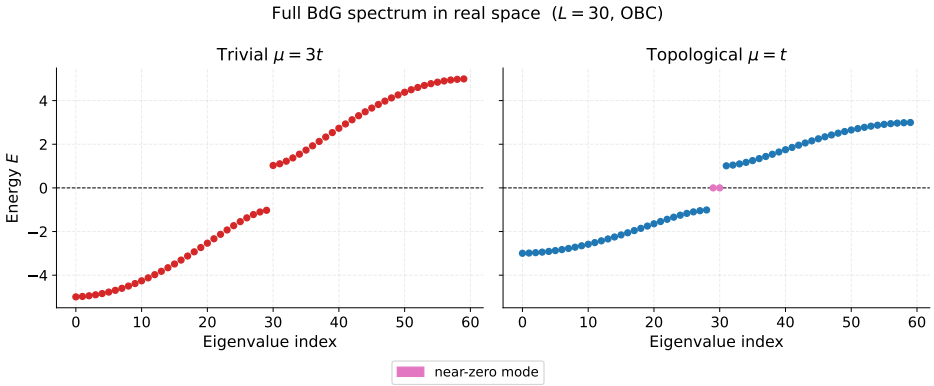
```
# Majorana splitting scan over chain lengths:
E0_arr = np.array([
    np.sort(np.abs(KitaevChain(L=l, ...).spectrum()))[0]
    for l in L_arr
])
```

Finite-size spectrum (OBC): edge modes appear



- Pink curve: near-zero mode pinned inside the gap across the entire topological phase $|\mu| < 2t$.
- Gray curves: bulk quasiparticle bands — gapped in both phases, closing only at $\mu = \pm 2t$.

Real-space BdG snapshot



Left (trivial, $\mu = -3t$): all eigenvalues uniformly gapped — no special states.
Right (topological, $\mu = -t$): two eigenvalues pinned at $E \approx 0$ — the Majorana pair localized at opposite ends of the chain.

Exponential localization of the edge mode

At the symmetric point $t = \Delta$, the zero-mode wavefunction has an exact analytical form. Substituting the Ansatz $\psi_j \sim \lambda^j$ into the bulk BdG equations gives:

$$\lambda = \frac{\mu}{2t}, \qquad |\lambda| < 1 \text{ (topological phase)}$$

The left edge mode decays as:

$$\psi_j^L \propto \left(\frac{\mu}{2t}\right)^{j-1} = e^{-(j-1)/\xi}$$

with coherence length

$$\xi = \frac{1}{\ln(2t/|\mu|)}$$

Key points

- Pure exponential at $t = \Delta$ (no oscillations)
- $\xi \to \infty$ as $|\mu| \to 2t$ (phase boundary)
- $\xi \to 0$ as $|\mu| \to 0$ (sweet spot)

Derivation: finite-size splitting $E_0(L)$

In a **finite** chain, the left and right edge modes overlap, lifting the degeneracy. First-order perturbation theory gives:

$$\delta E = \frac{|\langle \psi^L | H | \psi^R \rangle|}{\|\psi^L\| \, \|\psi^R\|}$$

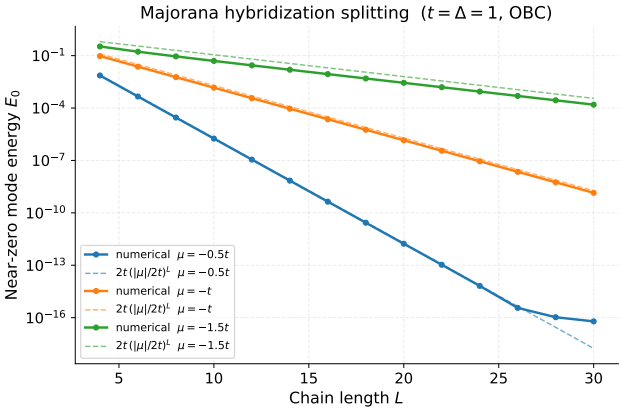
The overlap is dominated by the mid-chain region:

$$\langle \psi^L | \psi^R \rangle \sim \sum_{j=1}^{L} \lambda^{j-1} \cdot \lambda^{L-j} = L \, \lambda^{L-1}$$

Absorbing normalisation and subleading factors into a prefactor fixed by the exact sweet-spot calculation:

$$\boxed{E_0(L) \approx 2t \left(\frac{|\mu|}{2t} \right)^L}$$

Majorana hybridization splitting vs system size



Key result — at the symmetric point $t = \Delta$:

$$E_0(L) \approx 2t\left(\frac{|\mu|}{2t}\right)^L$$

- Straight lines on the semilog scale confirm **exponential** decay.
- Slope $= \ln(|\mu|/2t) = -1/\xi$ measures the **coherence length** ξ.
- Smaller $|\mu| \Rightarrow$ faster decay $\Rightarrow$ better protected zero mode.

Block 2 — key takeaways

1. The BdG matrix is built directly from μ, t, Δ — exact diagonalization gives the full spectrum with no approximations.
2. OBC spectra show **near-zero edge modes** in the topological phase — direct numerical confirmation of bulk-boundary correspondence.
3. Edge modes are **exponentially localized** with coherence length $\xi = 1/\ln(2t/|\mu|)$.
4. Majorana splitting decays as $E_0 \sim (|\mu|/2t)^L$ — exponential in L, measurable from the semilog slope.

Next (Block 3): Jordan–Wigner transform $\rightarrow$ qubit encoding $\rightarrow$ quantum circuit for ground-state preparation.